Data dictionary

1. Dataset overview

File	Grain (one row per…)	Rows	Key	Period
beach_monthly.csv	Beach × month	9,600 (80 beaches × 120 months)	beach_nm + month	09/2016 – 08/2026
incidents.csv	Person involved in an incident	33,941 people in 32,459 incidents	incident_id + person_no	01/09/2016 – 31/08/2026
pricing.csv	Purchasable initiative / cost item	10	item	Current prices

2. General conventions

- **Dates.** Stored as text: individual dates as dd/mm/yyyy, months as mm/yyyy.
- **Children.** People aged under 18.
- **Local.** A resident of the same Local Government Area (LGA) as the beach. Everyone else (intrastate, interstate, international) is non-local. This definition applies to both tourist_ratio and local_ind.
- **Beach names.** Beach names are unique. Where Fryland has more than one beach with the same name, the suburb is added in brackets.

3. beach_monthly.csv

One row per beach per month, covering 80 Fryland ocean beaches across Sydney, Central Coast & Hunter, North Coast, Illawarra and South Coast from September 2016 to August 2026. Contains visitation, environmental conditions, beach characteristics and the level of each safety initiative in place. The "Varies" column shows how often a value can change for a given beach.

Column	Description	Type	Units / format	Varies
beach_nm	Beach name. Unique across Fryland; duplicate real names are disambiguated with the suburb, e.g. "Shelly Beach (Manly)".	chr	Text	Static
postcode	Fryland postcode of the beach. Several postcodes contain 2–4 beaches; program costs are charged per postcode.	int	4-digit code	Static

month	Calendar month of the record.	chr	mm/yyyy	–
visitors	Total beach visits in the month.	int	Visits	Monthly
shore_len_m	Shoreline length of the beach.	int	Metres	Static
n_rips	Long-term average number of rip currents at the beach.	dbl	Count, 1 dp	Static
air_temp_hi	Average daily maximum air temperature over the month.	dbl	°C	Monthly
swell_avg_m	Average swell height over the month.	dbl	Metres	Monthly
swell_max_m	Maximum swell height recorded in the month. Always ≥ swell_avg_m.	dbl	Metres	Monthly
wind_days	Number of high-wind days in the month.	int	Days	Monthly
water_temp	Average sea surface temperature over the month.	dbl	°C	Monthly
rain_mm	Total rainfall in the month.	dbl	mm	Monthly
turbidity	Turbidity index. Higher values mean poorer water clarity.	dbl	Index 0–100	Monthly
tourist_ratio	Share of visitors who are non-local (i.e. not residents of the beach's LGA; includes intrastate, interstate and international visitors).	dbl	Proportion 0–1	Monthly
lg_hrs	Lifeguard person-hours on patrol at the beach in the month.	dbl	Person-hours	Monthly
drone_hrs	Drone surveillance flight hours at the beach in the month.	dbl	Flight hours	Monthly
campaign_idx	Intensity of any active beach safety education campaign. 0 = no campaign.	dbl	Index 0–1	Monthly; same for all beaches in a postcode
first_aid_cap	Bystander / community first aid and rescue training coverage of the beachgoing public. This is not a measure of lifeguard competency; on-duty lifeguards are assumed to be first-aid qualified whenever patrolled = 1.	dbl	Index 0–100	Monthly; set at postcode level
pa_system	A public address (PA) system is installed at the beach.	int	1 = yes, 0 = no	Mostly static

flags	Permanent rip-warning flags / signage installed at the beach.	int	1 = yes, 0 = no	Mostly static
patrolled	The beach had any patrol in the month. Exactly equal to lg_hrs > 0.	int	1 = yes, 0 = no	Monthly
equip_rating	Rescue equipment effectiveness rating.	int	Rating 0–100	Updated yearly (July)
rocky	A rocky headland or rock platform is present.	int	1 = yes, 0 = no	Static
fishing	The beach is a popular fishing spot.	int	1 = yes, 0 = no	Static
n_bins	Number of rubbish bins at the beach.	int	Count	Near-static
rockpool	An ocean rockpool is present.	int	1 = yes, 0 = no	Static
icecream_visits	Average number of ice-cream truck visits per week during the month.	dbl	Visits/week, 1 dp	Monthly
surfer_prop	Share of visitors who are surfers.	dbl	Proportion 0–1	Monthly
marine_health	Marine health index category.	chr	L = low, M = medium, H = high	Monthly

4. incidents.csv

One row per person involved in a beach incident, for incidents occurring between 01 September 2016 and 31 August 2026. Most incidents involve one person; some involve two or three, sharing the same incident_id. The “Level” column shows whether a value is shared by everyone in the incident (Incident) or recorded separately for each person (Person).

Column	Description	Type	Units / format	Level
incident_id	Identifier of an incident event, numbered sequentially by date and hour of occurrence. An incident may involve 1–3 people.	int	Integer ID	Incident
person_no	Person number within the incident. (incident_id, person_no) uniquely identifies each row.	int	1–3	Person

dte_reported	Date the incident was reported. Never earlier than dte_occurred; usually the same day.	chr	dd/mm/yyyy	Incident
dte_occurred	Date the incident occurred.	chr	dd/mm/yyyy	Incident
inc_hour	Hour of the day the incident occurred (24-hour clock).	int	Hour 0–23	Incident
inc_type	Outcome for the person. Ordered Mild < CloseCall < Severe, where Severe = drowning death.	chr	Category	Person
inc_cause	Cause of the incident. NA = cause unspecified (occurs for some child rows only).	chr	Category	Incident (may be NA for individual children)
age	Age of the person. Children are people aged under 18.	int	Years	Person
sex	Sex of the person.	chr	M = male, F = female	Person
surfer	The person was surfing / is a surfer.	int	1 = yes, 0 = no	Person
swim_level	Self-reported or assessed swimming ability.	chr	Ordered category	Person
local_ind	Local status: local = resident of the same LGA as the beach; not_local = anyone else (intrastate, interstate or international).	chr	Category	Person
response_min	Minutes from occurrence of the incident to arrival of the first responder. Includes any delay in the incident being reported.	dbl	Minutes, 1 dp	Incident
beach_nm	Beach where the incident occurred. Matches beach_nm in beach_monthly.csv.	chr	Text	Incident
n_lifeguards	Number of lifeguards on duty at the time and place of the incident. 0 when the beach was not on patrol.	int	Count	Incident
drone_present	A drone was flying over the beach at the time of the incident.	chr	Y = yes, N = no	Incident
alone	The person was alone at the beach. Can only be 1 for single-person incidents.	int	1 = yes, 0 = no	Person

5. pricing.csv

One row per purchasable initiative or cost component. Use this table to cost a proposed allocation against the FD$2,000,000 per year budget. The linked_var column names the beach_monthly.csv column each item changes.

Column	Description	Type	Units / format	Values in file	Notes
Item	Name of the purchasable beach safety initiative or cost component.	chr	Text	10 items (e.g. Casual SLSC lifeguard, Drone hardware, PA system)	One row per item
Unit	Unit to which the cost applies.	chr	Text	e.g. per hour, per drone, per postcode per summer, per beach	Read with cost_fd
cost_fd	Cost per unit.	int	Fryland dollars ($FD)	450 – 80,000	For PA system, the value is the install cost; ongoing maintenance is described in notes
cost_type	How the cost is incurred.	chr	Category	one_off / hourly / annual / one_off + annual	one_off costs recur only when assets are replaced (e.g. drones: 3-year life)
linked_var	Column in beach_monthly.csv that the item changes. Links the price list to the modelling data.	chr	Column name	drone_hrs, lg_hrs, campaign_idx, first_aid_cap, pa_system, flags	
effect_per_unit	What one unit of the item buys, expressed as its change to linked_var.	chr	Text	e.g. +1 lg_hrs; Sets flags = 1; +10 first_aid_cap capped at 100	
notes	Additional conditions, prerequisites or timing (e.g. operator must be trained before flying; training takes effect from the following month).	chr	Text	Free text	

6. Tips on linking the datasets

- **incidents → beach_monthly:** Join on beach_nm and on the month derived from dte_occurred (convert dd/mm/yyyy to mm/yyyy). Every incident beach and month exists in beach_monthly.csv.
- **pricing → beach_monthly:** linked_var matches a column name in beach_monthly.csv, showing which variable an initiative would change.
- **Postcode-level costs:** Campaign and community first aid training are charged per postcode, and campaign_idx is identical for every beach in a postcode-month.
- **Building counts:** Incident counts per beach-month can be built by aggregating incidents.csv (count distinct incident_id for events, or rows for people) and joining to beach_monthly.csv. Beach-months with no incidents should be counted as zero.